

KUSH PATEL

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Education

Carnegie Mellon University - GPA: 3.73/4.0

Pittsburgh, PA

Master of Science in Artificial Intelligence Engineering

May 2027

CourseWork: AI Systems & Toolchains, Machine Learning, Advanced NLP (PhD), Deep Learning, ML with Large Datasets, Trustworthy AI, Distributed Systems

Rutgers University - New Brunswick - GPA: 3.67/4.0

Bachelor of Science in Computer Science & Minor in Business Administration

Sep 2022 – Aug 2024

CourseWork: Computer Algorithms, Artificial Intelligence, Deep Learning, Operating Systems, Systems Programming

Technical Skills

Languages: Python, Go, C, TypeScript, SQL

AI/ML: LLMs, RAG, AI Agents, MCP, Transformers, QLoRA, Quantization, OCR, Computer Vision

Tools: FastAPI, Flask, Docker, Kubernetes, Git, CI/CD, AWS, GCP, Azure

Data & Systems: Apache Spark, Kafka, PostgreSQL, BigQuery, dbt, FAISS, Polars

Frameworks: PyTorch, TensorFlow, Hugging Face, JAX, LangChain, LangGraph, OpenCV, ONNX, YOLO

Experience

Advantest America, Inc

South Plainfield, NJ

Software Engineer

Jun 2024 – Present

- Built a multi-agent AI system that reads team emails and attachments through an LLM parsing pipeline and updates version-controlled project files, **achieving 94% extraction accuracy**
- Developed an OCR inspection pipeline that tracks PCB layer stack-up and self-calibrates to each job size, **saving an estimated \$1M+ annually** in scrap and rework
- Designed ETL pipelines using Apache Spark and dbt with CI/CD-automated testing to clean and transform **730M+ records**, powering production reporting and analytics at 99% uptime
- Automated 80% of the inspection workflow** by connecting robotic arms to legacy measurement software via a TCP/IP communication bridge
- Engineered and deployed a scalable FastAPI server with caching and optimized data processing, powering a real-time dashboard **across 5 regions** for live manufacturing data

Technology Intern

Aug 2022 – May 2024

- Secured access for **500+ users** by configuring Azure AD IAM, MFA, and conditional access policies
- Automated audits and backup verification across **1,000+ endpoints**, streamlining compliance reporting

Projects and Research

Efficient LLM Reasoning with Early Exit | CMU | NLP Research | [Report](#)

2026

- Trained an MLP probe on 4,096-dim Llama-3.1-8B hidden states to predict answer readiness mid-reasoning (**0.82 ROC-AUC**), a lightweight alternative to reward models
- Reduced simulated inference tokens by **up to 98.1%** with a step-level early-exit policy, outperforming token-probability baselines on four benchmarks

Prompt Injection Defense for LLM Agents | CMU | LLM Security, AI Agents | [Report](#)

2026

- Built a 3-layer guardrail pipeline for LLM agents using structural delimiters, a router LLM firewall, and an LLM-as-a-judge, **cutting prompt injection success on InjecAgent from 75.0% to 2.3%**
- Added a deterministic symbolic verifier and TF-IDF RAG threat intelligence to catch attacks the LLM judge missed, **intercepting 97.7% of residual failures**

RAG Question-Answering System | CMU | Retrieval, Re-ranking, Web Crawling | [GitHub](#)

2026

- Developed a data pipeline to crawl, clean, and chunk web pages on CMU and Pittsburgh culture into a searchable knowledge base for question answering
- Improved F1 by 10% over BM25** by fusing dense MiniLM/FAISS retrieval with BM25 via RRF and adding cross-encoder re-ranking to ground Mistral-7B